

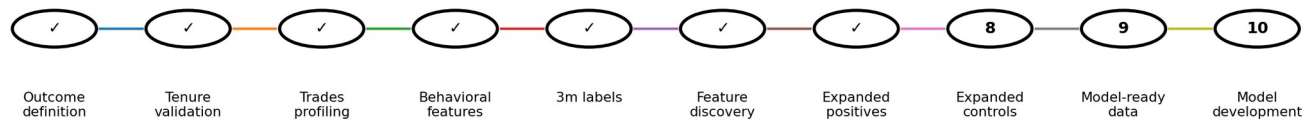
Fraud Alert Vulnerability - Technical Methodology and Results

Methodology, decisions, data quality findings, feature discovery, and current modeling preparation

Technical Confluence version | Current through expanded positive cohort

English + Español

CURRENT NEXT STEP → Expanded controls



Project progression and current next step. / Progresión del proyecto y siguiente paso actual.

1. Objective and modeling target / Objetivo y target de modelado

EN Predict propensity to place a first observed qualifying fraud alert, not confirmed fraud victimization.

ES Predecir propensión a colocar una primera alerta de fraude observada válida, no victimización de fraude confirmada.

EN Qualifying alert types currently used for outcome construction: type 2 (Initial Fraud Alert) and type 3 (Extended Fraud Alert), pending final business confirmation.

ES Tipos de alerta actualmente usados para construir el outcome: tipo 2 (Initial Fraud Alert) y tipo 3 (Extended Fraud Alert), pendiente confirmación final de negocio.

EN Primary discovery horizon: first observed qualifying alert within the next 3 calendar months after the monthly observation date.

ES Horizonte principal de discovery: primera alerta observada válida dentro de los siguientes 3 meses calendario después de la fecha mensual de observación.

KEY TAKEAWAY / MENSAJE CLAVE

EN: Point-in-time discipline is mandatory: no post-observation information is allowed in features.

ES: La disciplina point-in-time es obligatoria: no se permite información posterior a la observación en las features.

2. Alert history and recurrence review / Historia de alertas y recurrencia

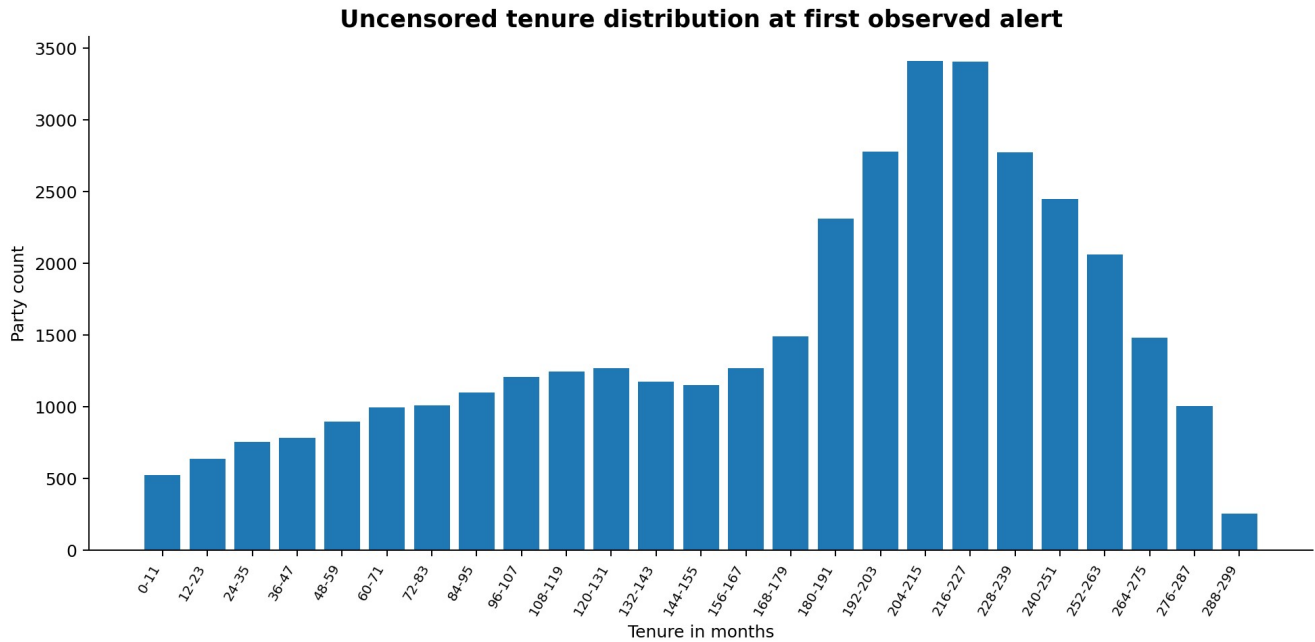
Metric	Result
security_alert source rows	324,268,295
Approx. parties	31,916,677
Min creation date	2007-01-13
Max creation date	2026-07-30
Unique creation-date events	42,318,245
Parties with alert	29,766,492
Creation within 7 years	10,951,174
Creation after 7 years	1,600,579

EN Different creation dates in the historical snapshots do not always represent a truly new alert; the earliest observed qualifying alert is therefore used as the modeling anchor.

ES Fechas de creación diferentes en los snapshots históricos no siempre representan una alerta realmente nueva; por ello se utiliza la primera alerta válida observada como ancla de modelado.

3. Tenure construction and censoring / Construcción de tenure y censura

Metric	Result
Relationship-history source	bc3.person_restricted
Relationship start definition	MIN(data_date)
Global history minimum	2000-09-30
Left-censored sample parties	29,086 / 66,643 (43.64%)
Uncensored valid parties	37,492
Median tenure	195 months
Average tenure	175.23 months
P90 tenure	257 months



Uncensored valid tenure distribution. The peak shifts by alert year, indicating a cohort effect rather than a causal tenure-risk peak. / Distribución de tenure válida no censurada. El pico cambia por año de alerta, indicando un efecto de cohorte más que un pico causal de riesgo.

EN Tenure is retained as contextual information, but it should not be labeled “high risk” without a denominator population.

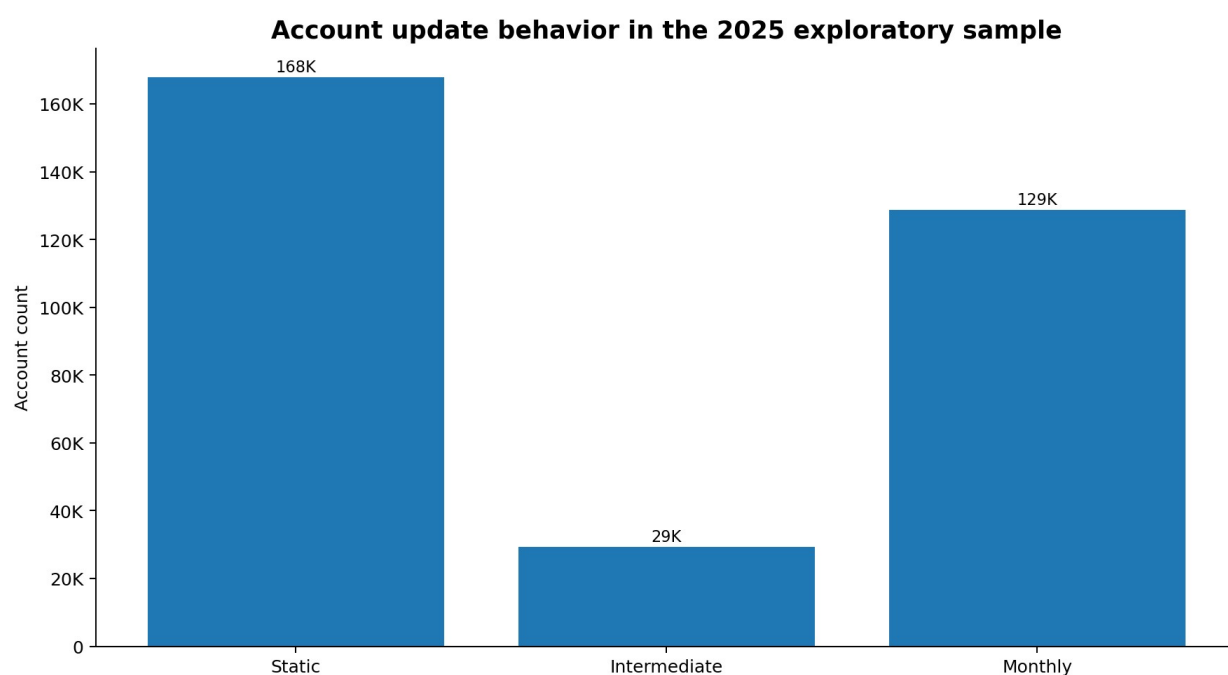
ES Tenure se conserva como información contextual, pero no debe etiquetarse como “alto riesgo” sin una población denominador.

4. Trades source profiling / Perfilamiento de trades

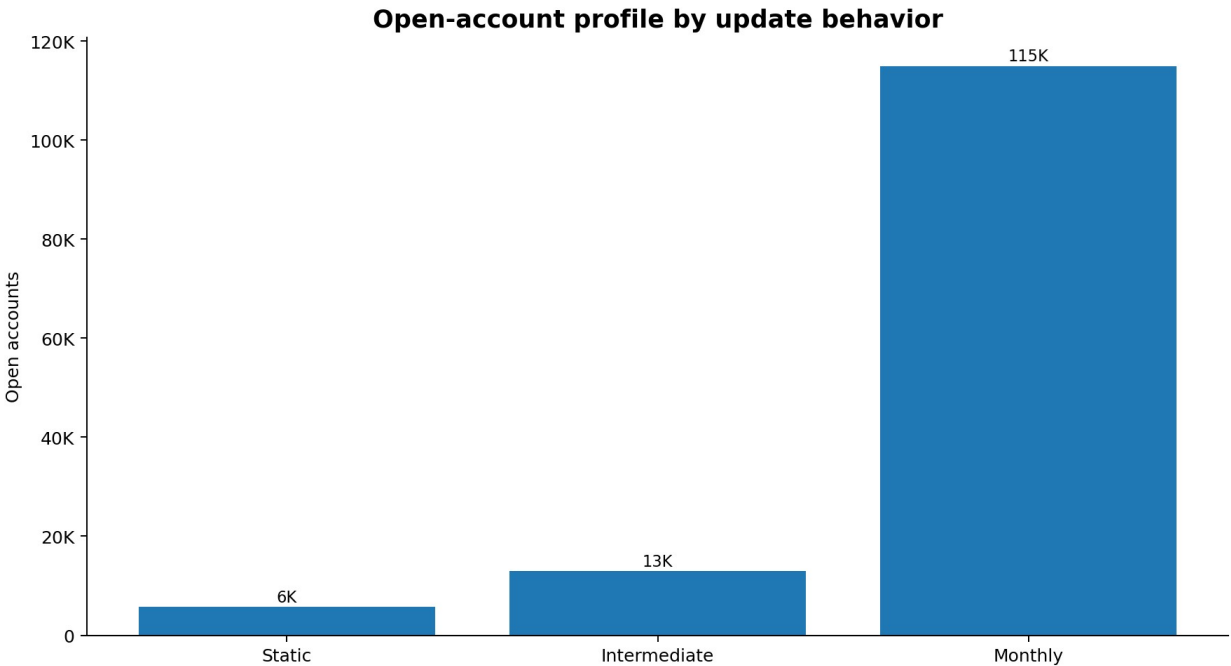
Metric	Result
Source scale	~504.8B rows
Recent monthly rows	~3.26B
Recent monthly parties	~302M
Missing eff_dte	0
eff_dte after data_date	20 rows
Sample duplicate key party_id+acct_id+data_date	0
Median staleness (data_date-eff_dte)	275 days

EN data_date is the load/availability date; eff_dte is the last verification date from the data provider. Staleness therefore reflects information freshness, not ingestion lag.

ES data_date es la fecha de carga/disponibilidad; eff_dte es la última fecha de verificación del proveedor. Por ello, staleness refleja frescura de información y no retraso de ingestión.



Account update behavior was segmented only to understand lifecycle/reporting patterns, not as a direct risk feature. / El comportamiento de actualización se segmentó solo para entender lifecycle/reporting, no como feature directa de riesgo.



Among open accounts, monthly reporting dominates, reinforcing that raw update frequency is largely a lifecycle/reporting-state indicator. / Entre cuentas abiertas domina el reporte mensual, reforzando que la frecuencia cruda de actualización refleja principalmente lifecycle/reporting.

5. MOP transition logic / Lógica de transición MOP

EN MOP codes 01-06 are treated as ordinal delinquency levels. 09/9B/8A/8B are active severe-adverse states; 9P/8P are resolved historical-adverse states; UR is handled separately. Codes 07 and 08 remain unknown and are not inferred.

ES Los códigos MOP 01-06 se tratan como niveles ordinales de morosidad. 09/9B/8A/8B son estados adversos severos activos; 9P/8P son estados adversos históricos resueltos; UR se maneja por separado. Los códigos 07 y 08 permanecen desconocidos y no se infieren.

Transition metric	2025 result
No change	3,501,681 (98.79%)
Deterioration	29,310 (0.83%)
Improvement	7,843 (0.22%)
New adverse status (initial EDA)	4,042 (0.11%)
Validated DPD deterioration flag	29,310
Entered active adverse flag	3,394
Adverse resolved flag	477

EN The key behavioral features use true effective updates and lagged account status, not repeated monthly snapshots alone.

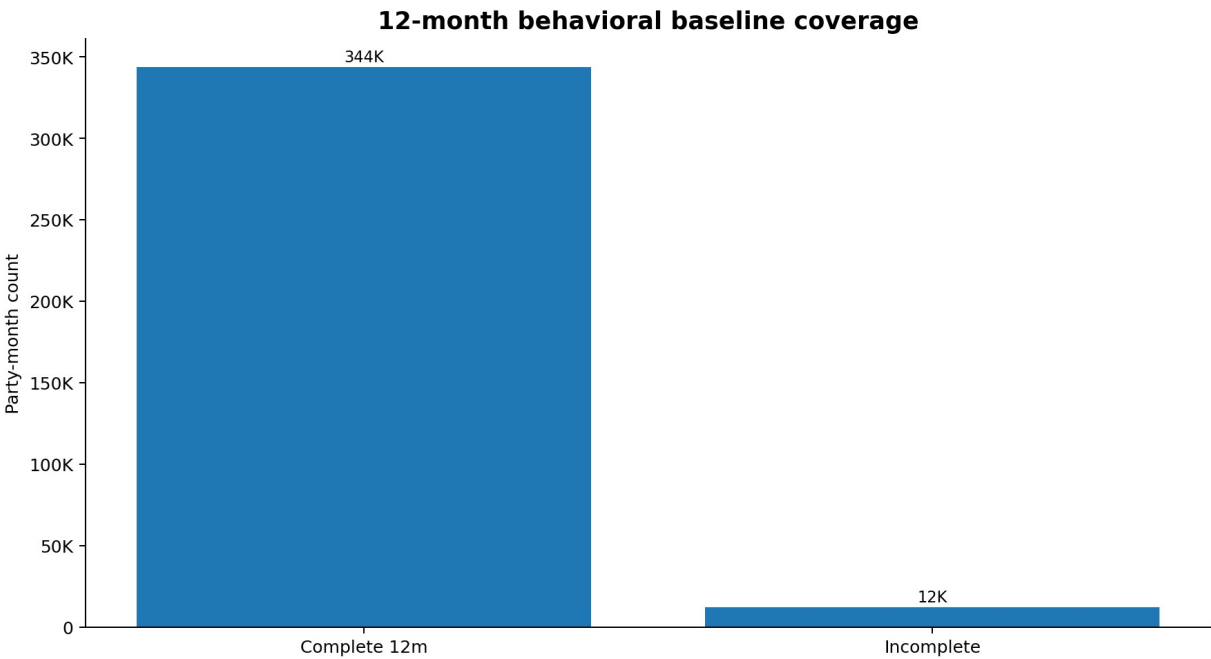
ES Las features conductuales clave utilizan actualizaciones efectivas reales y estatus rezagado de la cuenta, no únicamente snapshots mensuales repetidos.

6. Behavioral feature architecture / Arquitectura de features conductuales

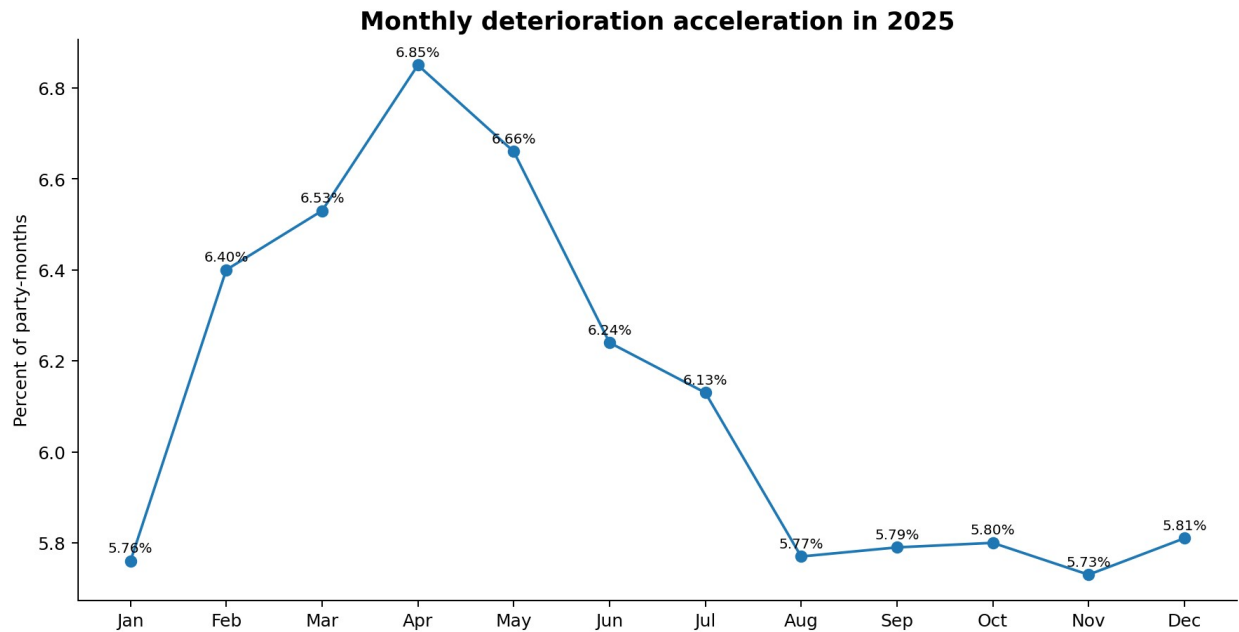
EN Features compare a recent 3-month window against a non-overlapping prior 9-month baseline.

ES Las features comparan una ventana reciente de 3 meses contra un baseline previo no traslapado de 9 meses.

Feature family	Examples
Deterioration	deteriorations_recent_3m; deterioration_acceleration; max_dpd_jump_recent_3m
Adverse events	adverse_recent_3m; adverse_acceleration
Delinquency pressure	past_due_change_vs_baseline; delinquency_share_change_vs_baseline
Exposure pressure	balance_change_vs_baseline; utilization_change_vs_baseline
Context/exposure	active accounts and recent/prior account averages

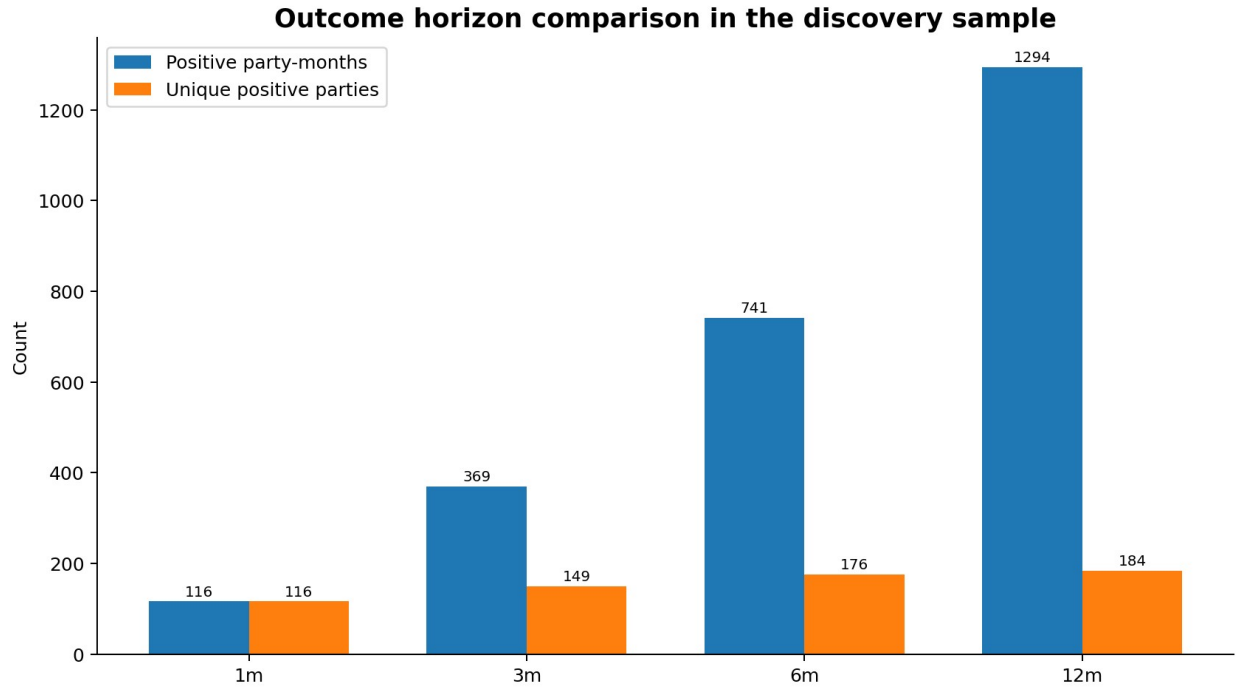


Complete 12-month baseline coverage is ~96.6%, supporting stable rolling feature construction. / La cobertura de baseline completo de 12 meses es ~96.6%, soportando una construcción estable de features móviles.



Monthly deterioration acceleration is reasonably stable across 2025, with a modest Q1/Q2 increase. / La aceleración mensual del deterioro es razonablemente estable durante 2025, con un incremento moderado en Q1/Q2.

7. Outcome labels and horizon selection / Labels de outcome y selección de horizonte



Party-month positives and unique positive parties in the initial 2025 discovery sample. / Party-months positivos y personas positivas únicas en la muestra inicial de discovery 2025.

Horizon	Positive party-months	Unique positive parties	Interpretation
1m	116	116	Too sparse for primary development
3m	369	149	Primary discovery horizon
6m	741	176	Secondary benchmark
12m	1,294	184	~50% censoring in 2025 sample; benchmark only

EN Positive counts at 3m and 6m are party-months, so a single future alert can label several prior monthly observations. For case-control feature discovery, one first-positive observation per person was selected.

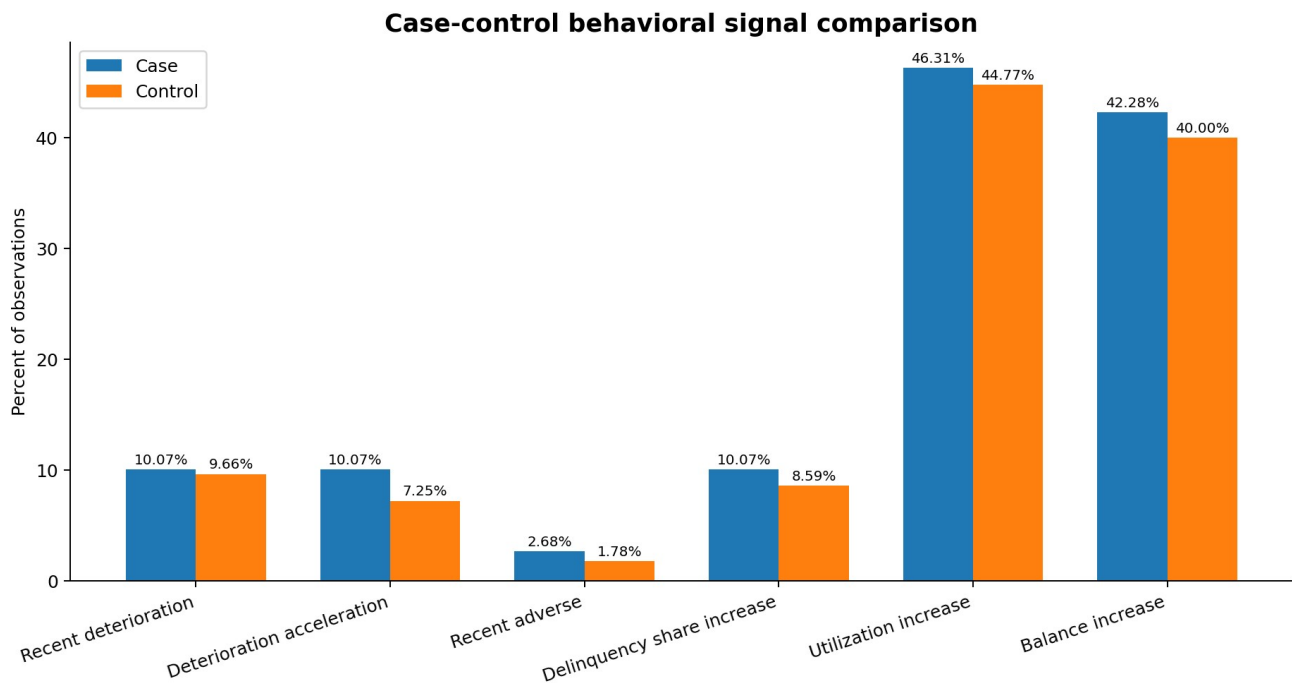
ES Los conteos positivos a 3m y 6m son party-months, por lo que una sola alerta futura puede etiquetar varias observaciones mensuales previas. Para feature discovery case-control se seleccionó una sola primera observación positiva por persona.

8. Matched case-control discovery design / Diseño de discovery case-control emparejado

Metric	Cases	Controls
Rows	149	2,980
Unique parties	149	2,784
Avg active accounts	7.22	7.21
Median active accounts	6	6
P90 active accounts	14	15

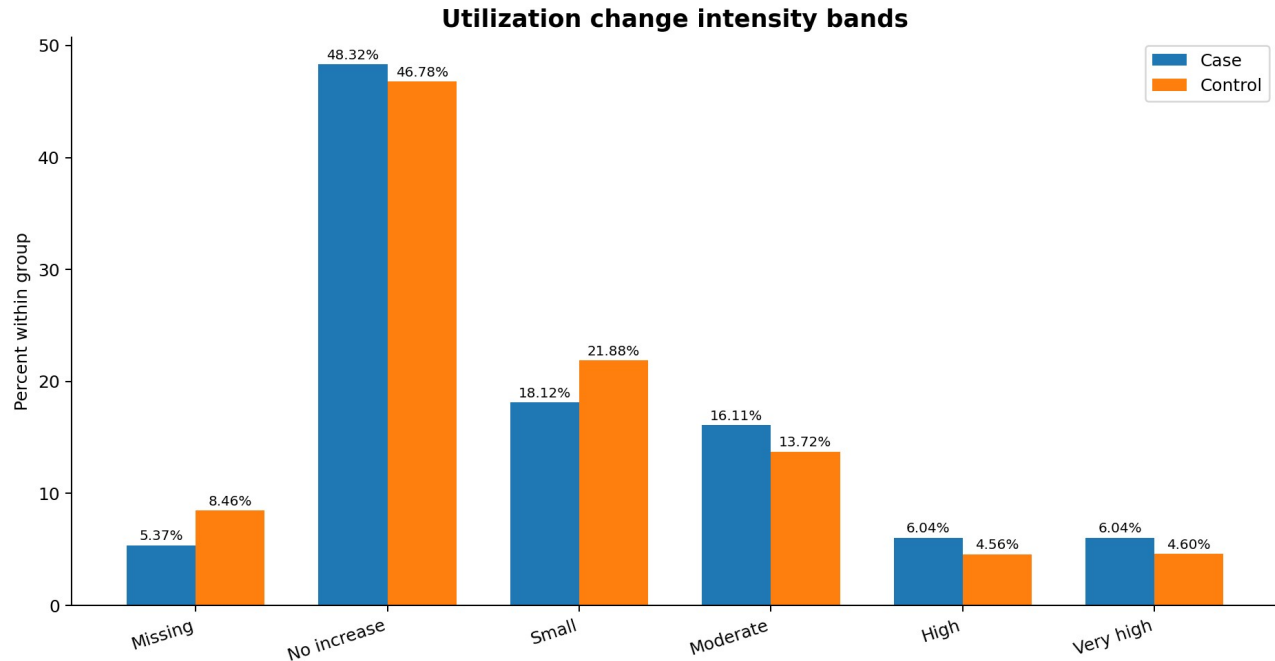
EN Controls were matched within the same observation month and active-account band. The design was used for exploratory feature discovery, not for calibrated population risk estimation.

ES Los controles se emparejaron dentro del mismo mes de observación y banda de cuentas activas. El diseño se utilizó para feature discovery exploratorio, no para estimación calibrada de riesgo poblacional.

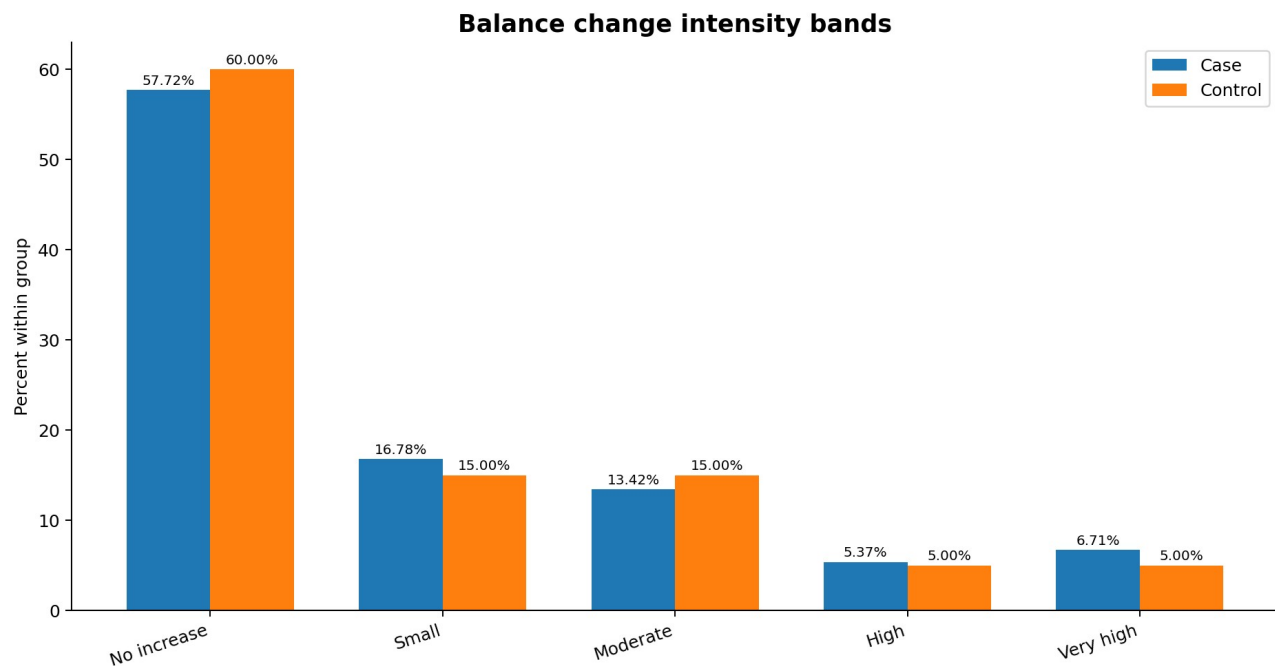


Binary signal prevalence. Deterioration acceleration separates better than raw recent deterioration. / Prevalencia de señales binarias. Deterioration acceleration separa mejor que recent deterioration crudo.

9. Signal intensity findings / Hallazgos de intensidad de señales



Utilization-change bands based on exploratory control percentiles. Moderate/high/very-high bands are relatively more represented among cases. / Bandas de cambio de utilización basadas en percentiles exploratorios de controles. Las bandas moderate/high/very-high están relativamente más representadas en casos.

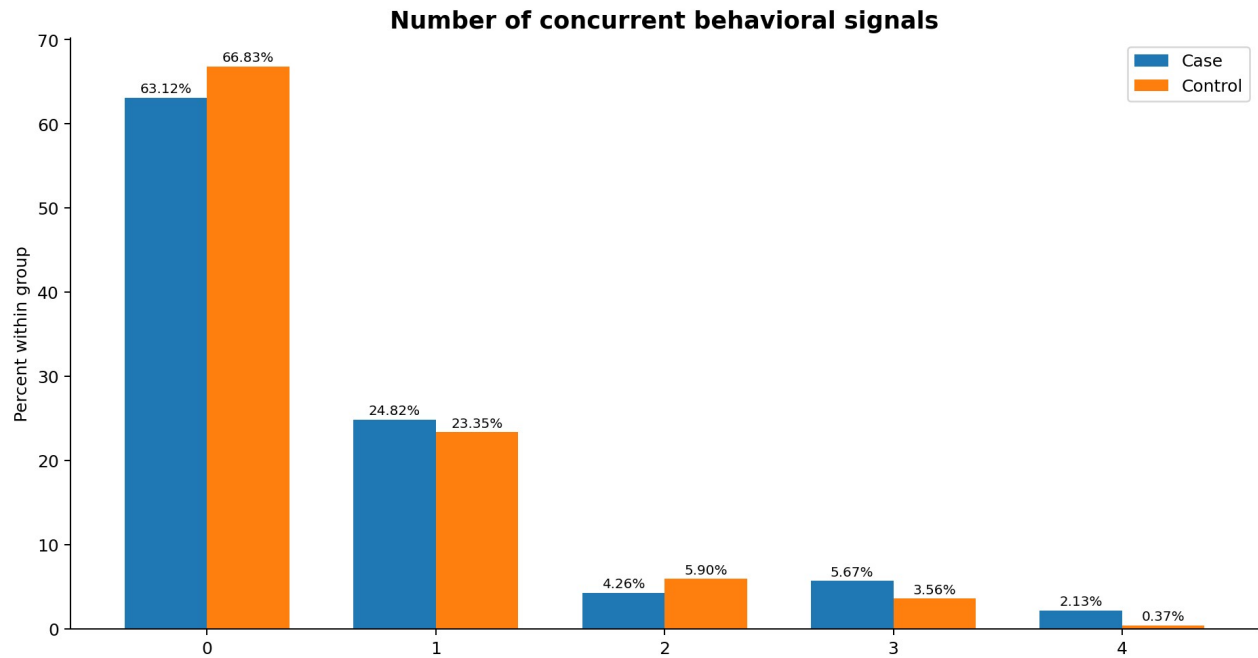


Balance-change signal is weaker and concentrated mainly in the most extreme band. / La señal de cambio de balance es más débil y se concentra principalmente en la banda más extrema.

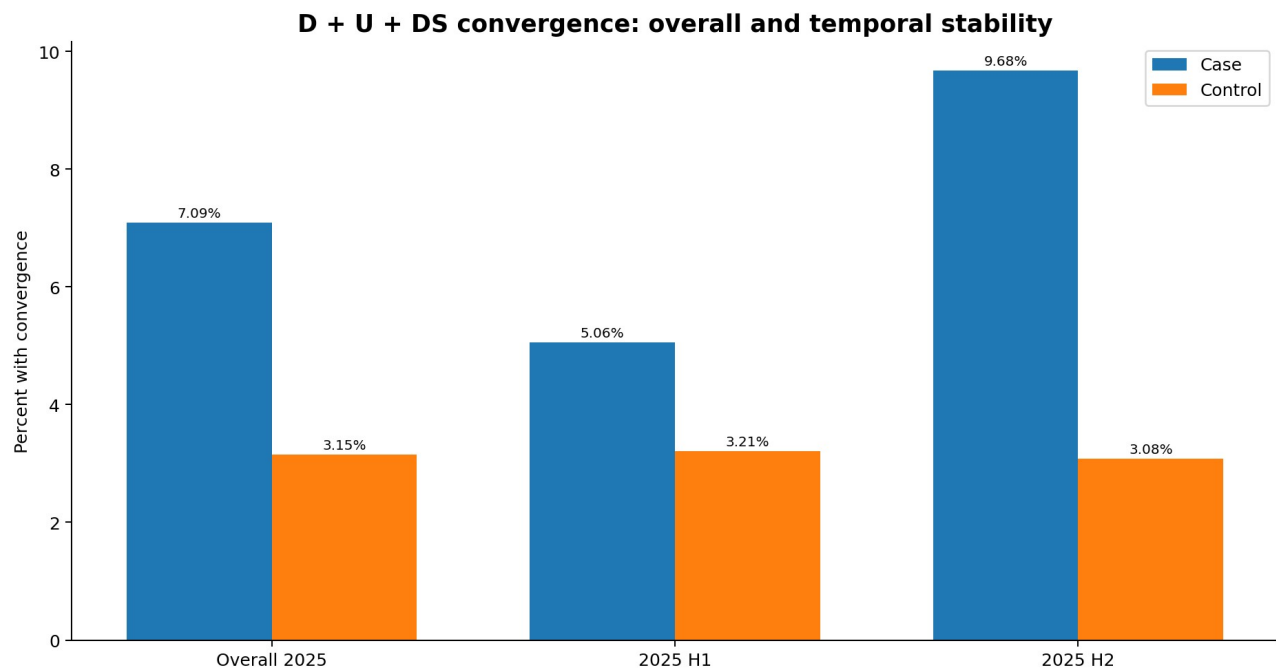
EN The EDA suggests that magnitude of change versus personal baseline contains more information than a simple increase/no-increase flag.

ES El EDA sugiere que la magnitud del cambio contra el baseline personal contiene más información que un simple flag de aumento/no aumento.

10. Signal convergence / Convergencia de señales



A simple equal-weight signal count is not monotonic; the identity of the combined signals matters. / Un conteo simple de señales con igual peso no es monótonico; importa qué señales se combinan.



D+U+DS is the most promising exploratory interaction: deterioration acceleration + utilization pressure + delinquency-share increase. / D+U+DS es la interacción exploratoria más prometedora: aceleración del deterioro + presión de utilización + aumento de delinquency share.

Period	Case convergence	Control convergence	Relative prevalence
Overall 2025	7.09%	3.15%	2.25x
2025 H1	5.06%	3.21%	1.58x
2025 H2	9.68%	3.08%	3.14x

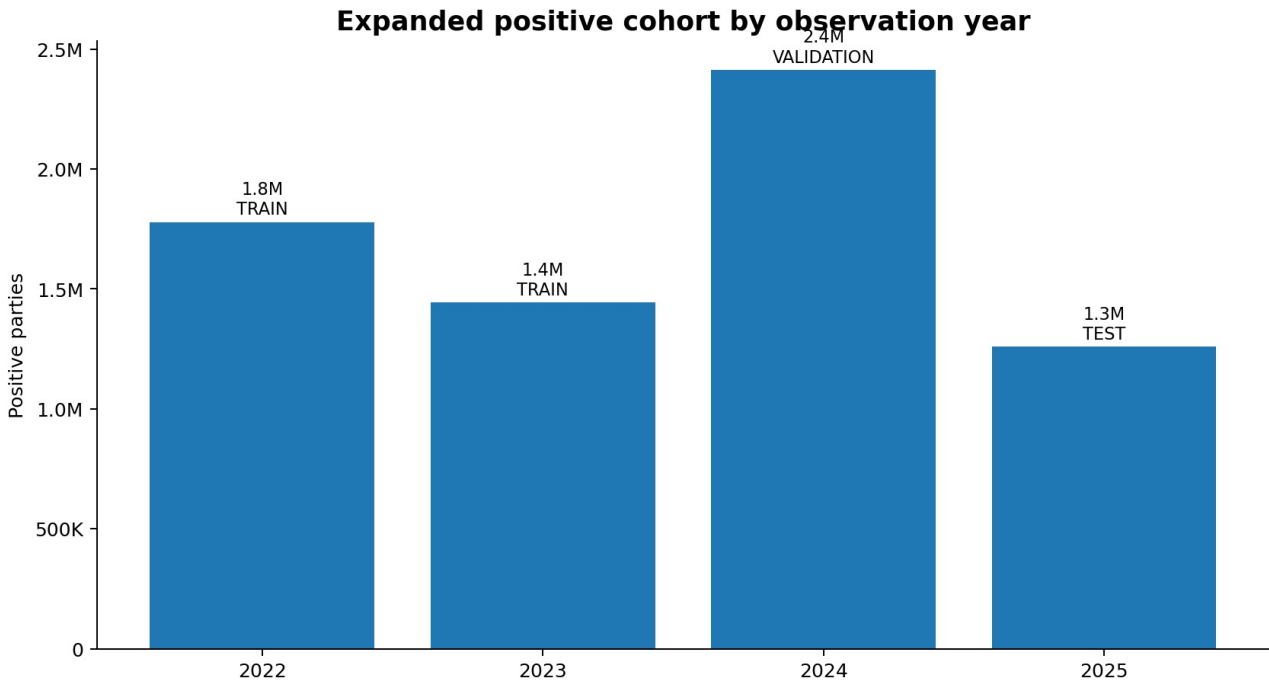
EN The direction is consistent in H1 and H2, but only 10 total cases in the discovery sample carry the D+U+DS pattern. It is a hypothesis to validate at scale, not a production rule.

ES La dirección es consistente en H1 y H2, pero solo 10 casos totales en la muestra de discovery presentan el patrón D+U+DS. Es una hipótesis para validar a escala, no una regla productiva.

11. Expanded positive cohort / Cohorte positiva ampliada

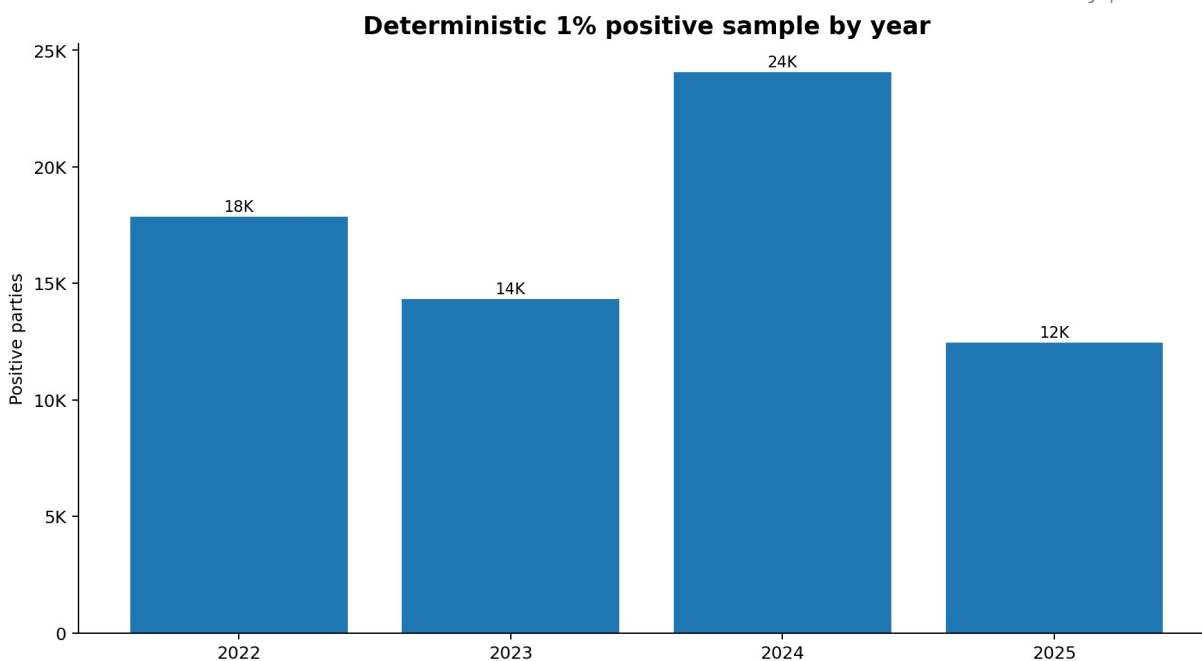
EN The modeling observation period was expanded to 2022-2025. Positive anchors use a 3-calendar-month horizon and are assigned by observation year.

ES El período de observación para modelado se amplió a 2022-2025. Los positive anchors usan un horizonte de 3 meses calendario y se asignan por año de observación.



Chronological split: 2022-2023 TRAIN, 2024 VALIDATION, 2025 TEST. / Split cronológico: 2022-2023 TRAIN, 2024 VALIDATION, 2025 TEST.

Status check	Count
Alert on/before observation after boundary anchoring	22,511,193
Valid positive anchor	6,894,117
After 3m window	0



Deterministic 1% positive sample = 68,733 parties, designed to validate features at much larger scale without processing all positives immediately. / Muestra positiva determinística de 1% = 68,733 personas, diseñada para validar features a mayor escala sin procesar inmediatamente todos los positivos.

12. Key methodological decisions / Decisiones metodológicas clave

EN Model target is first observed qualifying alert, not confirmed fraud loss.

ES El target es la primera alerta válida observada, no pérdida de fraude confirmada.

EN Use point-in-time observation dates and calendar-month label horizons.

ES Usar fechas de observación point-in-time y horizontes de label en meses calendario.

EN Separate tenure censoring from descriptive analysis.

ES Separar censura de tenure del análisis descriptivo.

EN Do not use raw account update frequency as a direct risk signal.

ES No usar frecuencia cruda de actualización de cuentas como señal directa de riesgo.

EN Keep continuous behavior-change features; EDA thresholds are not frozen production cutoffs.

ES Conservar features continuas de cambio conductual; los thresholds de EDA no son cortes productivos congelados.

EN Use chronological model splits and keep each party within a single split.

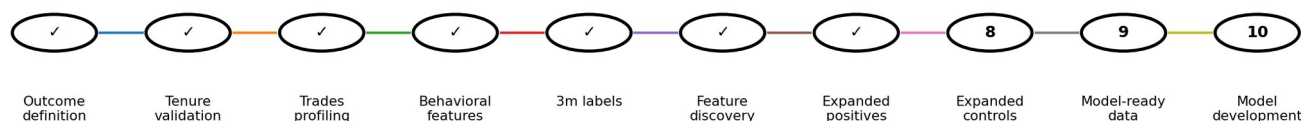
ES Usar splits cronológicos y mantener cada party dentro de un solo split.

EN Case-control samples require later weighting/calibration to recover population prevalence.

ES Las muestras case-control requieren ponderación/calibración posterior para recuperar prevalencia poblacional.

13. Current roadmap position / Posición actual del roadmap

CURRENT NEXT STEP → Expanded controls



Completed through deterministic expanded-positive sampling. The immediate next milestone is the expanded control cohort. / Completado hasta el muestreo determinístico de positivos ampliados. El siguiente hito inmediato es la cohorte ampliada de controles.

Roadmap item	Status
18. Expanded positive cohort 2022-2025	COMPLETE
19. Temporal TRAIN/VALIDATION/TEST split	COMPLETE
20. Deterministic expanded positive sample	COMPLETE
21. Expanded control cohort	NEXT
22. Filter trades to modeling cohort	PENDING
23. Rebuild point-in-time features at scale	PENDING
24. Out-of-sample signal validation	PENDING
25+. Modeling, calibration, governance	PENDING

14. Immediate next steps / Sigüientes pasos inmediatos

- EN** Build the expanded control cohort with eligibility rules comparable to positives.
- ES** Construir la cohorte ampliada de controles con reglas de elegibilidad comparables a positivos.
- EN** Prevent party leakage across TRAIN, VALIDATION, and TEST.
- ES** Evitar leakage de party entre TRAIN, VALIDATION y TEST.
- EN** Join/filter trades only for selected modeling parties and required lookback dates.
- ES** Unir/filtrar trades únicamente para las personas seleccionadas y fechas de lookback requeridas.
- EN** Rebuild recent-3m/prior-9m features point-in-time.
- ES** Reconstruir features recent-3m/prior-9m point-in-time.
- EN** Validate D/U/DS and other discovery findings on expanded TRAIN, then freeze transformations and test on 2024/2025.
- ES** Validar D/U/DS y otros hallazgos de discovery en TRAIN ampliado, luego congelar transformaciones y probar en 2024/2025.
- EN** Develop baseline models, evaluate rare-event metrics, calibrate, and document governance.
- ES** Desarrollar modelos baseline, evaluar métricas de evento raro, calibrar y documentar gobernanza.